

Low- Q point-addition circuit b6f2b0a

Source-grounded anatomy of submission b6f2b0a / commit e6d9fa9

Submitted operating point

$Q = 825$ logical qubits
 $T = 489,161,900$ Toffoli gates
 $Q \times T = 403,558,567,500$

Inversion architecture

Register-shared, shrunken Proos–Zalka-style EEA
1,479 scheduled reversible steps
No dialog transcript and no dialog-GCD replay

Q825-specific change

Remove one persistent ℓ_q lane
Reconstruct its high bit implicitly
Use an eight-scratch split-five kernel

What this circuit optimizes. The benchmark circuit maps a quantum accumulator $R = (x_R, y_R)$ and a classically specified addend $A = (x_A, y_A)$ to $R' = R + A$ over `secp256k1`. The low- Q route spends substantial nonlinear work to minimize the maximum number of simultaneously live logical qubits.

Evidential status. An isolated replay of all 9,024 challenge tests reports zero classical, phase, and ancilla failures. The seven-bit quotient-length promise is tied to that finite validation support; the deck does not promote it to an all-input proof or a full windowed-Shor resource estimate.

The Q825 circuit is analyzed in an isolated restoration and is separate from the 8e9c9a2 product-score deck.

What “Shrunken-PZ” means

A low-width reversible organization of Proos–Zalka-style modular inversion

1. Mathematical task

For $x \not\equiv 0 \pmod{p}$, an extended Euclidean computation produces a coefficient t such that

$$tx + sp = \gcd(x, p) = 1, \quad t \equiv x^{-1} \pmod{p}.$$

The point-addition circuit uses this map to compute $\lambda = d_y d_x^{-1} \bmod p$.

2. Why it is “shrunken”

During the Euclidean recurrence, the remainders become shorter while the Bézout coefficients grow. Rather than reserve separate full-width registers for every quantity, the implementation tracks their current lengths and packs nonoverlapping values into shared fixed-width banks.

3. The implementation used here

Submission **b6f2b0a** uses the register-shared descendant of this architecture: two 259-qubit work banks, compact length/control metadata, and a fixed schedule of 1,479 reversible steps. Dynamic cursors address only the active packed fields.

Packed state in b6f2b0a. One bank contains the evolving coefficient, quotient, and remainder (t, q, r) ; the other contains (t', r') . Their source-enforced live-width bounds are

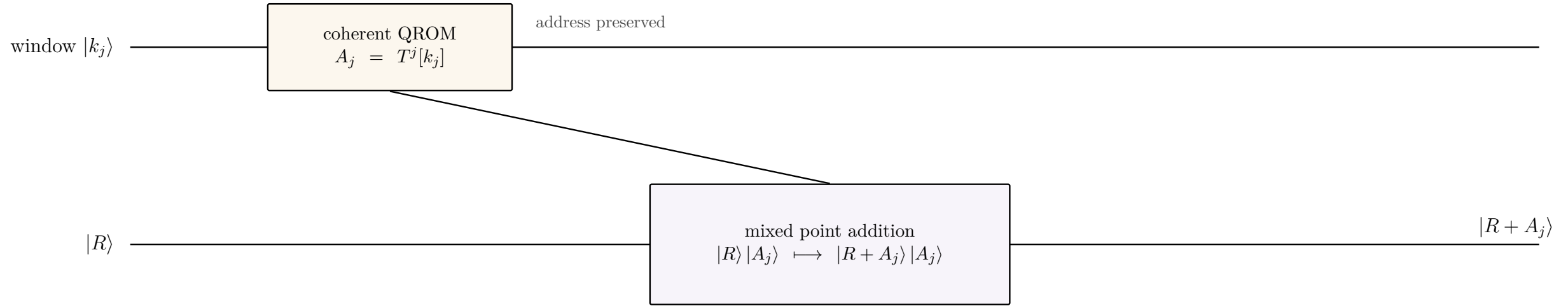
$$\text{bitlen}(t) + 1 + \text{bitlen}(q) + \text{bitlen}(r) \leq 259, \quad \text{bitlen}(t') + \text{bitlen}(r') \leq 259.$$

Reusing the space vacated by shrinking values is the central width-saving mechanism.

Not dialog-GCD. “Shrunken-PZ” is project shorthand for this low-width Proos–Zalka-style EEA lineage. The Q825 circuit carries the complete packed Euclidean state and reverses the scheduled updates to clean it; it does *not* store a compressed branch transcript for later dialog-GCD replay.

Where the low- Q kernel sits in windowed Shor

The submitted source implements one mixed addition; coherent point selection is outside its interface.



Challenge interface

x_A and y_A arrive as classical bits. The circuit may materialize them transiently, use classical-control specialization, and release them before the inversion peak.

Full windowed interface

A_j is selected coherently by k_j . Its coordinates are quantum data entangled with the address until QROM unlookup, so classical-only savings do not transfer automatically.

Interpretation

The deck explains the submitted point-addition artifact. A complete Shor integration requires a separately validated lookup/use/unlookup construction and compatible accounting.

Affine mixed addition implemented by b6f2b0a

$R = (x_R, y_R)$, $A = (x_A, y_A)$, and $R' = R + A$ in the generic case

In-place register updates

Input: $X = x_R$, $Y = y_R$; classical addend $A = (x_A, y_A)$; $x_R \neq x_A$.

1. $Y \leftarrow y_A - Y$ $\triangleright Y = d_y = y_A - y_R$
2. $X \leftarrow x_A - X$ $\triangleright X = d_x = x_A - x_R$
3. $\lambda \leftarrow YX^{-1}$ $\triangleright \lambda = d_y/d_x$
4. $Y \leftarrow 0$ $\triangleright$ reuse Y as scratch
5. $X \leftarrow \lambda^2 - x_R - x_A$ $\triangleright X = x_{R'}$
6. $X \leftarrow x_A - X$ $\triangleright X = d'_x = x_A - x_{R'}$
7. $Y \leftarrow \lambda X$ $\triangleright Y = d'_y = \lambda(x_A - x_{R'})$
8. $\lambda \leftarrow 0$ $\triangleright \lambda = d'_y/d'_x$ witnesses cleanup
9. $X \leftarrow x_A - X$; $Y \leftarrow Y - y_A$ $\triangleright X = x_{R'}, Y = y_{R'}$

Slope convention

The source uses

$$d_x = x_A - x_R, \quad d_y = y_A - y_R, \quad \lambda = d_y d_x^{-1}.$$

Reversing both differences leaves the usual slope unchanged.

Coordinate identities

$$x_{R'} = \lambda^2 - x_R - x_A, \quad y_{R'} = \lambda(x_A - x_{R'}) - y_A.$$

The implementation overwrites X and Y rather than allocating separate output coordinates.

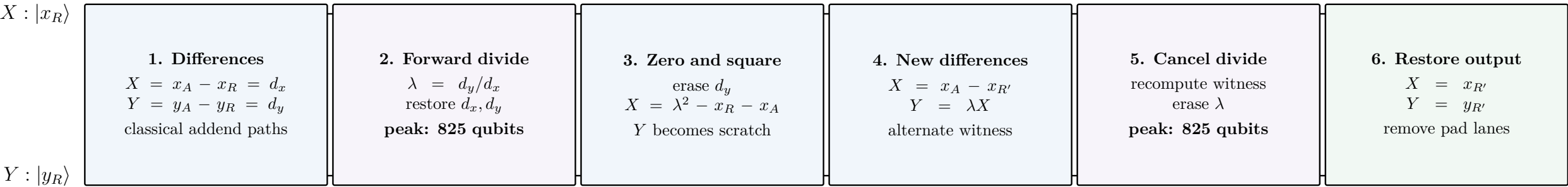
Two division lifecycles

The first EEA traversal produces λ from (d_x, d_y) .

After the coordinates are formed, the identity $\lambda = d'_y/d'_x$ permits a second EEA traversal to erase λ without retaining the first inversion history.

Point-addition pipeline and source correspondence

Both 825-qubit peaks occur inside the register-shared EEA stages.



Source sections and live-width maxima					
Differences	Forward divide	Zero and square	New differences	Cancel divide	Restore output
ec3.dy_build, ec3.dx_build	ec3.inv_fwd	ec3.dy_zero, ec3.new_x	ec3.alt.new_dx, new_dy	ec3.alt.cancel	ec3.alt.new_x_restore, new_y_restore
max $Q = 788$	max $Q = 825$	max $Q = 789$	max $Q = 788$	max $Q = 825$	max $Q = 789$

Count-only source instrumentation reproduces the submitted Q and structural T exactly.

Why the register-shared core peaks at 825 qubits in Stage 2 and 5

The Q825 route removes one persistent metadata lane; local microkernels then consume the remaining 13-qubit headroom.

Persistent state during a scheduled EEA step

W_1 259 lanes	W_2 259 lanes	passenger d_y 257 lanes	metadata 37
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The two physical banks: 518 qubits

W_1 is newly allocated. W_2 is not another copy of d_x : the 257-lane divisor register receives two clean pad lanes and is reversed in place to become the second 259-lane packed bank.

Passenger coordinate: 257 qubits

The numerator d_y remains live while the EEA acts on d_x . Its canonical top lane is known to be zero and is borrowed by selected local kernels, but it still belongs to the passenger register and is restored before terminal multiplication.

Persistent metadata: 37 qubits

$9_{\ell_t} + 1_{\ell_{t'}} + 7_{\ell_q} + 8_{\ell_s} + 8_{\ell_{r'}} + 4_{\text{control}} = 37.$

The four controls are the two phase bits, sign, and iteration parity.

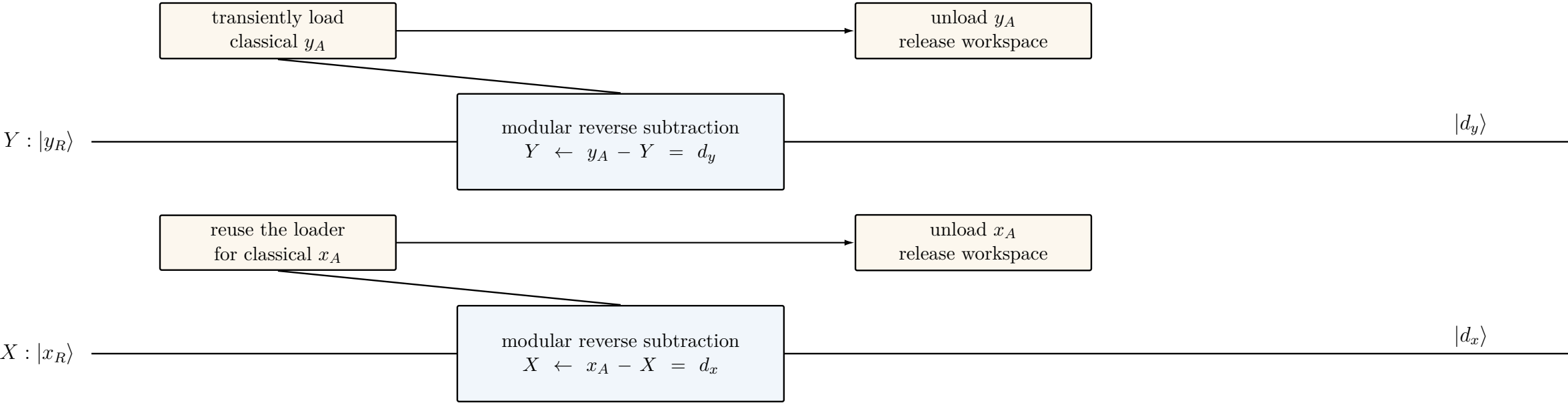
Peak equation.

$$\underbrace{259 + 259 + 257 + 37}_{812 \text{ persistent qubits}} + \underbrace{13}_{\text{largest scheduled scratch set}} = 825.$$

The Q826 parent stored an eight-bit ℓ_q , giving $813 + 13 = 826$. Q825 keeps only seven persistent ℓ_q lanes and reconstructs the omitted controlled effect when required.

Stage 1: build d_y and d_x with low-pressure classical loads

$d_y = y_A - y_R \pmod{p}$, followed by $d_x = x_A - x_R \pmod{p}$



Serialized materialization

The two 256-bit classical coordinates are not simultaneously materialized. A transient quantum loader is used for one subtraction, restored to zero, and then reused.

Pseudo-Mersenne arithmetic

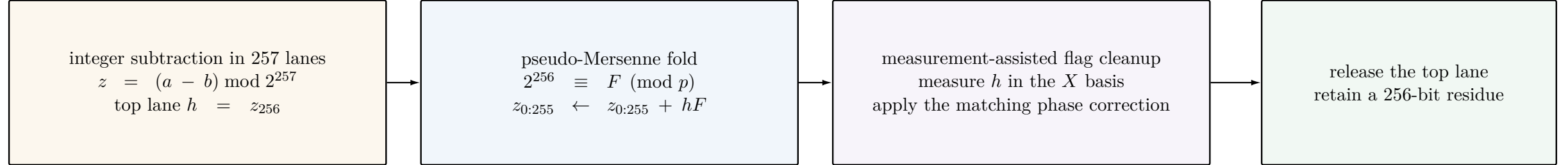
With $p = 2^{256} - (2^{32} + 977)$, overflow and borrow corrections are folded through the small constant $2^{32} + 977$ instead of a generic full-width modular reduction.

Scope

These loads exploit the challenge's classical-addend interface. In a coherent QROM-based Shor call, x_A and y_A are quantum data and need a different implementation.

Stage 1: fold the 257th lane instead of retaining a reducer

The point-addition route repeatedly trades canonical range for a narrow pseudo-Mersenne fold and prompt cleanup.



Sparse correction

$$p = 2^{256} - F, \quad F = 2^{32} + 977.$$

An overflow contribution $h2^{256}$ is congruent to hF . The correction therefore touches only the low window spanned by the sparse constant rather than retaining a second full-width modular-reduction register.

Range discipline

The raw fold produces a representative in $[0, 2^{256})$, not necessarily in $[0, p)$. Call sites that require a canonical value perform one final conditional subtraction of p ; paired inverse routines rely on the exact range contract established by the corresponding forward operation.

Approximation boundary

Several narrow comparisons and fold windows are selected for the benchmark route. The algebra $2^{256} \equiv F \pmod{p}$ is exact, while the claim that every omitted carry or comparison bit is irrelevant is supported only on the documented validation corpus unless separately proved.

The measurement-assisted cleanup itself is exact when its regenerated predicate matches the measured flag; the finite-support caveat concerns the selected arithmetic widths and range promises.

Stage 2: Algorithm behind shrunk-PZ modular inversion

Restoring long division inside a reversible extended-Euclidean schedule

Register-shared modular inverse

Input: $x \in \{1, \dots, p-1\}$ for the odd prime p . **Output:** $z = x^{-1} \bmod p$; all work registers are restored.

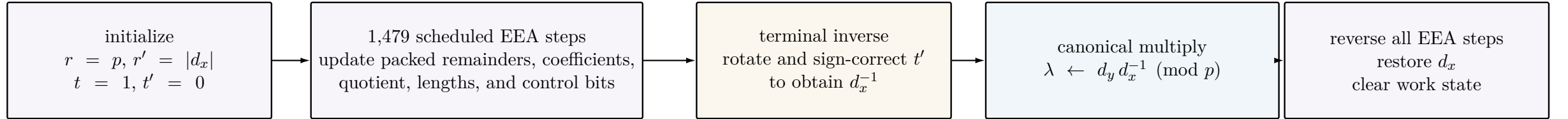
1. $x_0 \leftarrow \min(x, p-x); \quad \pi \leftarrow [x > p/2].$ $\triangleright x_0 \equiv (-1)^\pi x \pmod{p}$
2. $(t, r) \leftarrow (1, p); \quad (t', r') \leftarrow (0, x_0); \quad q \leftarrow 0.$ $\triangleright$ initialize the two packed Euclidean pairs
3. **while** $r' \neq 0$ **do** $\triangleright \gcd(r, r') = 1$
4. $q \leftarrow 0; \quad j \leftarrow \text{bitlen}(r) - \text{bitlen}(r').$ $\triangleright$ begin one restoring division
5. **for** $j, j-1, \dots, 0$ **do**
6. **if** $r \geq 2^j r'$ **then** $r \leftarrow r - 2^j r'; \quad q \leftarrow q + 2^j.$ $\triangleright q = \lfloor r_{\text{in}}/r' \rfloor$
7. $t' \leftarrow t' + qt.$ $\triangleright$ consume the same quotient bits in reverse order
8. $((t, r), (t', r')) \leftarrow ((t', r'), (t, r)); \quad \pi \leftarrow \pi \oplus 1.$ $\triangleright$ advance to the next Euclidean pair
9. Continue inactive padding until the fixed 1,479-microstep schedule ends. $\triangleright$ input-independent reversible runtime
10. $z \leftarrow (-1)^{\pi+1} t' \pmod{p}$; use z , then reverse Steps 2–9. $\triangleright r = 1, \quad r' = 0, \quad xz \equiv 1 \pmod{p}$

Euclidean invariants $rt + r't' = p,$
 $r \equiv (-1)^{\pi+1} xt', \quad r' \equiv (-1)^\pi xt \pmod{p}.$

Packed-state invariants $W_1 = t \| q \| r, \quad W_2 = \text{rot}_{\ell_s}(t' \| r'),$
 $\ell_t + 1 + \ell_q + \text{bitlen}(r) \leq n + 3, \quad \text{bitlen}(t') + \ell_{r'} \leq n + 3.$
 Here $n = 256$ is the `secp256k1` field-element bit width; rot_{ℓ_s} cyclically rotates the packed layout by ℓ_s lanes
 and is not an arithmetic shift.

Stage 2: register-shared shrunk-PZ division

A reversible extended Euclidean state machine computes $\lambda = d_y d_x^{-1}$ without a dialog transcript.



No recorded dialog

Dialog-GCD stores a compressed branch transcript and later replays it. Here the circuit carries the evolving Euclidean state itself in two packed work words and reverses the same scheduled gate sequence. The cost is high T ; the benefit is very low resident width.

Register-shared core

$$W_1 = (t \mid q \mid r), \quad W_2 = \text{rot}_{\ell_s}(t' \mid r').$$

Length metadata $\ell_t, \ell_q, \ell_{t'}, \ell_{r'}, \ell_s$ identifies the active subranges; phase, sign, and iteration bits control the reversible microsteps. The cyclic rotation aligns the active slices of t' and r' with the current arithmetic window without allocating shifted copies.

Passenger control

Only one full coordinate accompanies the EEA peak. The preserved clean top lane of the passenger is repeatedly borrowed as temporary storage and restored before the terminal multiplication or reverse schedule needs it.

Source: `register_shared_divide_forward`, `register_shared_forward`, and `register_shared_reverse`.

Stage 2: the packed Euclidean state machine

The circuit incrementally forms quotient bits and updates a remainder/coefficient pair.

Packed invariants

$$W_1 = t \parallel q \parallel r, \quad W_2 = \text{rot}_{\ell_s}(t' \parallel r').$$

For $n = 256$, each physical work word has $n + 3 = 259$ lanes, while

$$\ell_t + 1 + \ell_q + \text{bitlen}(r) \leq n + 3, \quad \text{bitlen}(t') + \ell_{r'} \leq n + 3.$$

One scheduled microstep

A shift counter ℓ_s selects an aligned slice of r' . The reversible core conditionally forms

$$r \leftarrow r \mp 2^{\ell_s} r', \quad t' \leftarrow t' \mp 2^{\ell_s} t,$$

toggles the corresponding quotient/sign information, and updates phase bits. When the quotient and shift counters return to zero, (r, t) and (r', t') are exchanged.

Fixed-length schedule

The state machine always executes 1,479 steps.

After the Euclidean computation reaches $r' = 0$, reversible padding steps preserve a fixed circuit shape. At the terminal boundary,

$$r = 1, \quad r' = 0, \quad q = 0,$$

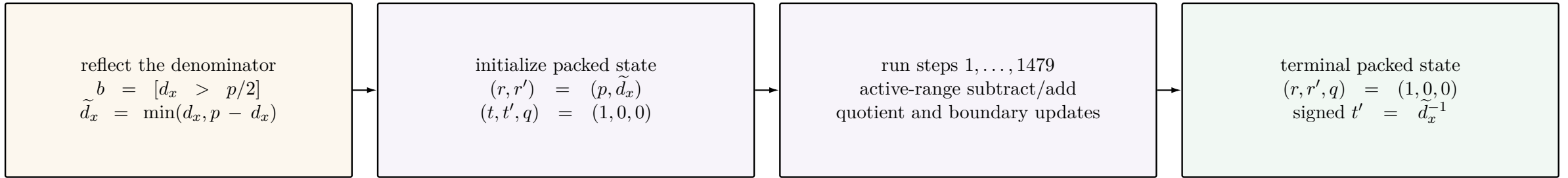
and the signed coefficient t' represents the modular inverse.

Why this differs from dialog-GCD. No compact branch word is stored for later interpretation. Instead, the active remainders, coefficients, quotient fragment, and their length metadata remain the reversible state. The inverse circuit is obtained by traversing the scheduled microsteps in reverse order.

The production route combines exact reversible microkernels with support-derived width schedules and metadata promises. Consequently, the architectural identities above are broader than the finite-support evidence for every selected truncation and hosting decision.

Stage 2: reflected input and a fixed reversible schedule

The circuit normalizes the denominator, runs a constant number of packed updates, and later reverses them gate for gate.



Why reflect first

Replacing d_x by $p - d_x$ when $d_x > p/2$ ensures $0 < \tilde{d}_x < p/2$. The initial remainder length therefore fits in eight metadata lanes after the source's mixed-width encoding. The parity bit b records the sign change needed to recover d_x^{-1} .

Padding after convergence

Inputs converge at different iterations, but a quantum circuit cannot branch into variable gate counts without recording that branch. The route instead executes a fixed 1,479-step schedule; once $r' = 0$, terminal-padding transitions preserve the terminal representation.

Exact restoration

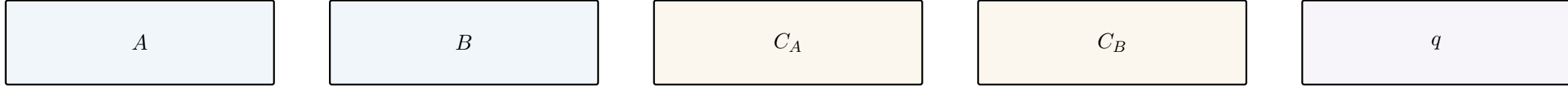
After the terminal coefficient has been used, constant terminal fields are rebuilt and the same 1,479 microsteps are applied in reverse order. This restores the padded d_x register, clears both work banks and all metadata, and finally undoes the initial reflection.

Unlike dialog-GCD, reversibility is carried by the complete packed Euclidean state and the inverse gate schedule; no branch transcript is retained.

Stage 2: register sharing and dynamic width

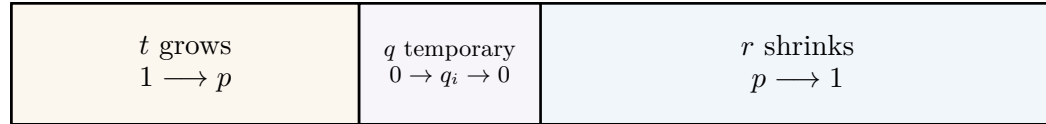
The principal low- Q gain comes from assigning physical lanes by lifetime rather than by logical register name.

Naive logical layout: five separate worst-case allocations



Production packed layout

W_1 (259 lanes)



W_2 (259 lanes, rotated by ℓ_s)



$x_0 = \min(x, p - x)$. At termination, $z = (-1)^{\pi+1} t'_{\text{term}} \bmod p = x^{-1} \bmod p$; each quotient q_i is assembled, consumed, and cleared within one restoring-division round.

Sliding ownership boundaries

As a remainder loses high bits, the paired coefficient acquires low-order space. Length registers move the logical boundary without keeping two independent 256-bit allocations.

Support-tuned schedule

Per-step active ranges, comparison widths, and quotient caps are selected from a thin schedule. High lanes known to be inactive are released or loaned to local predicates and carry chains.

Reversible lending discipline

A borrowed lane may host temporary information only while its logical owner is inactive. Each microkernel must restore the lane before the owner or a neighboring peak stage becomes live again.

Stage 2: dynamic cursors avoid copied position registers

Length metadata is temporarily mutated into scan state and then restored, saving a persistent cursor allocation.

Conventional reversible scan

$$|\ell_t\rangle |0\rangle_c \mapsto |\ell_t\rangle |\ell_t\rangle_c.$$

A copied cursor c is decremented while the active coefficient slice is traversed, then incremented and uncomputed. The copy is simple but remains live beside the packed data.

Production in-place scan

$$|\ell_t\rangle \xrightarrow{\text{scan}} |\ell_t - m\rangle \xrightarrow{\text{reverse scan}} |\ell_t\rangle.$$

The length register itself controls the active positions. Reverse traversal restores every decrement before another microstep may read the original length.

Coefficient traversal

The in-place ℓ_t cursor drives comparison, addition, and subtraction over the occupied portion of t . Only the local increment/decrement scratch is required; no second nine-lane copy of ℓ_t is retained.

Rotation cursor

The eight-lane ℓ_s register selects the cyclic alignment

$$W_2 = \text{rot}_{\ell_s}(t' \| r').$$

Shifted remainder and coefficient updates therefore read aligned physical slices without allocating shifted copies of either operand. This is a reversible permutation of storage lanes, not multiplication by 2^{ℓ_s} ; inverse rotation restores the original packed layout.

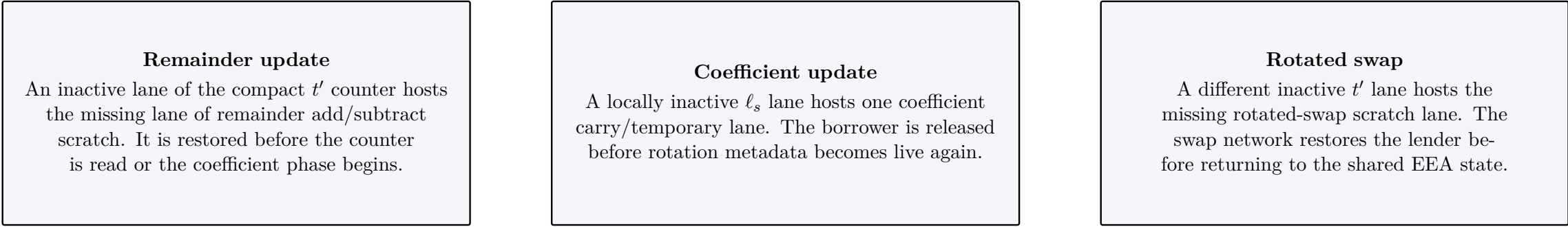
Lifecycle obligation

This is a coherent quantum cursor, not a classical loop index. The optimization is valid only because no concurrent operation needs the original metadata during the scan and the inverse traversal restores both the cursor and every borrowed scratch lane.

The dynamic-cursor mechanism is specific to the register-shared low- Q route; the $Q \times T$ circuit's ordinary generator indices do not occupy logical qubits.

Stage 2: three disjoint Q826 host windows inherited by Q825

Each window replaces one owned scratch lane by borrowing an inactive lane from the packed state.



Scheduled ownership



Why the savings compose. The three borrowers are never simultaneously active and do not borrow the same live value. Source-level harnesses compare each hosted kernel with an owned-scratch baseline in both directions, check scratch restoration, and verify that the host is clean at entry and exit.

Relation to Q825. These windows establish the Q826 peak with an eight-lane ℓ_q . Q825 keeps all three host schedules unchanged and removes one additional persistent lane from ℓ_q ; the next slide explains how that omitted control is reconstructed.

Stage 2: omit one persistent ℓ_q bit in the Q826 $\rightarrow$ Q825 transition

This submission saves one peak lane by reconstructing the high bit only where it is consumed.

Before: eight stored quotient-length lanes

$$\ell_q = (\ell_{q,7} \ell_{q,6} \cdots \ell_{q,0})_2.$$

The Q826 route keeps all eight bits resident while three disjoint host windows reuse other lanes. The persistent $\ell_{q,7}$ lane participates in the peak even though the submitted support never observes $\ell_q \geq 128$.

After: seven stored lanes plus implicit reconstruction

$$\ell_q = (\ell_{q,6} \cdots \ell_{q,0})_2, \quad \ell_{q,7} \text{ is not persistently allocated.}$$

When a split-five length kernel needs the omitted bit, it evaluates an equivalent zero-prefix predicate directly and writes only the final controlled effect.

Implicit high-bit identity

For a 259-bit source, let Z_k mean that every source bit in positions $k - 1$ and above is zero. The 2^7 bit of the source bit length is

$$h_7 = Z_{256} \oplus Z_{128}.$$

Thus the high bit can control a target without being materialized.

Eight-scratch split-five kernel

The prior serial split-five route used nine scratch lanes. Q825 materializes only length bits $2^4, 2^5, 2^6$ and reconstructs 2^7 inside a one-short subtraction/MCX path, reducing the qualified scratch requirement from nine lanes to eight.

Boolean factorization

Terms sharing h_7 are factored, for example

$$h_7 y \oplus h_7 b = h_7 (y \oplus b),$$

and reachable zero-prefix relations simplify the final-borrow controls. This replaces a stored bit with recomputation and additional multi-controlled operations.

The committed support audit reports $\max \ell_q = 22$ over 18,048 inversion factors. This supports the seven-bit route on the benchmark corpus; it is not, by itself, a universal proof that every field input satisfies $\ell_q < 128$.

Stage 2: consuming an omitted quotient-length bit

The eighth bit is represented by a reversible predicate only during the few kernels that need it.

1. Derive the logical bit from the source

Let $L = \text{bitlen}(w)$ for the current 259-bit source and define

$$Z_k(w) = \bigwedge_{i=k-1}^{258} \neg w_i.$$

Then $Z_{256} = [L \leq 255]$,
 $Z_{128} = [L \leq 127]$, and therefore

$$h_7 = [128 \leq L \leq 255] = Z_{256} \oplus Z_{128}.$$

2. Apply its effect without storing it

Kernels that would ordinarily read the resident lane $\ell_{q,7}$ instead compute the zero-prefix predicate, use it as a control, and immediately reverse the predicate network. For a target operation U , the logical action is

$$|w\rangle |\psi\rangle \mapsto |w\rangle U^{h_7(w)} |\psi\rangle,$$

with no persistent qubit allocated for h_7 .

3. Borrow only at an ownership gap

A direct metadata-swap subroutine temporarily places the logical high bit in a clean local lane denoted $zero_s$. The lane is borrowed only after its previous value is known to be zero and is restored to zero before the surrounding predicate or shift schedule can read it again.

Scratch reduction in the split-five kernel

The parent route materialized four length bits and required nine local scratch lanes. Q825 materializes only the 2^4 , 2^5 , and 2^6 bits; the implicit 2^7 effect is folded into the final subtraction and multi-controlled predicates, reducing the local requirement to eight lanes at the relevant substep.

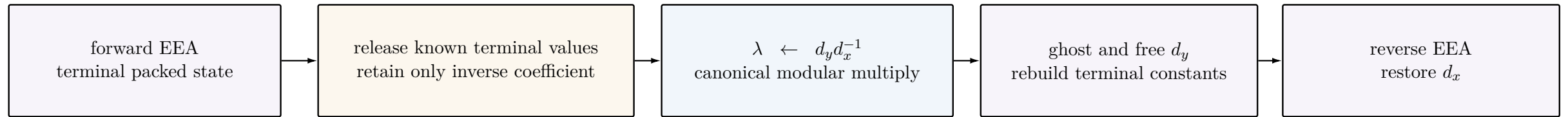
What is established

The source audit observes $\max L = 22$ over 18,048 inversion factors in the committed benchmark support, so the omitted bit is zero on those executions. The reversible predicate explains how the circuit remains coherent when that bit is consumed; the finite audit does not prove the same bound for every possible field input.

This is a recomputation-for-storage trade: one persistent peak lane is removed at the price of additional predicate and multi-controlled operations.

Stage 2: producing λ while preserving d_x and d_y

Terminal constants are released before multiplication, and the passenger is reconstructed after reverse EEA.



Terminal teardown

At convergence, most EEA fields are constants. The implementation toggles them to zero and releases their lanes before allocating the 257-lane λ result. This prevents the complete terminal EEA pack from overlapping the modular multiplication.

Passenger ghosting

After λ is formed, each d_y lane is Hadamard-measured and represented by a classical ghost record. The quantum lanes are freed while the EEA runs backward. The source later recomputes

$$d_y = \lambda d_x$$

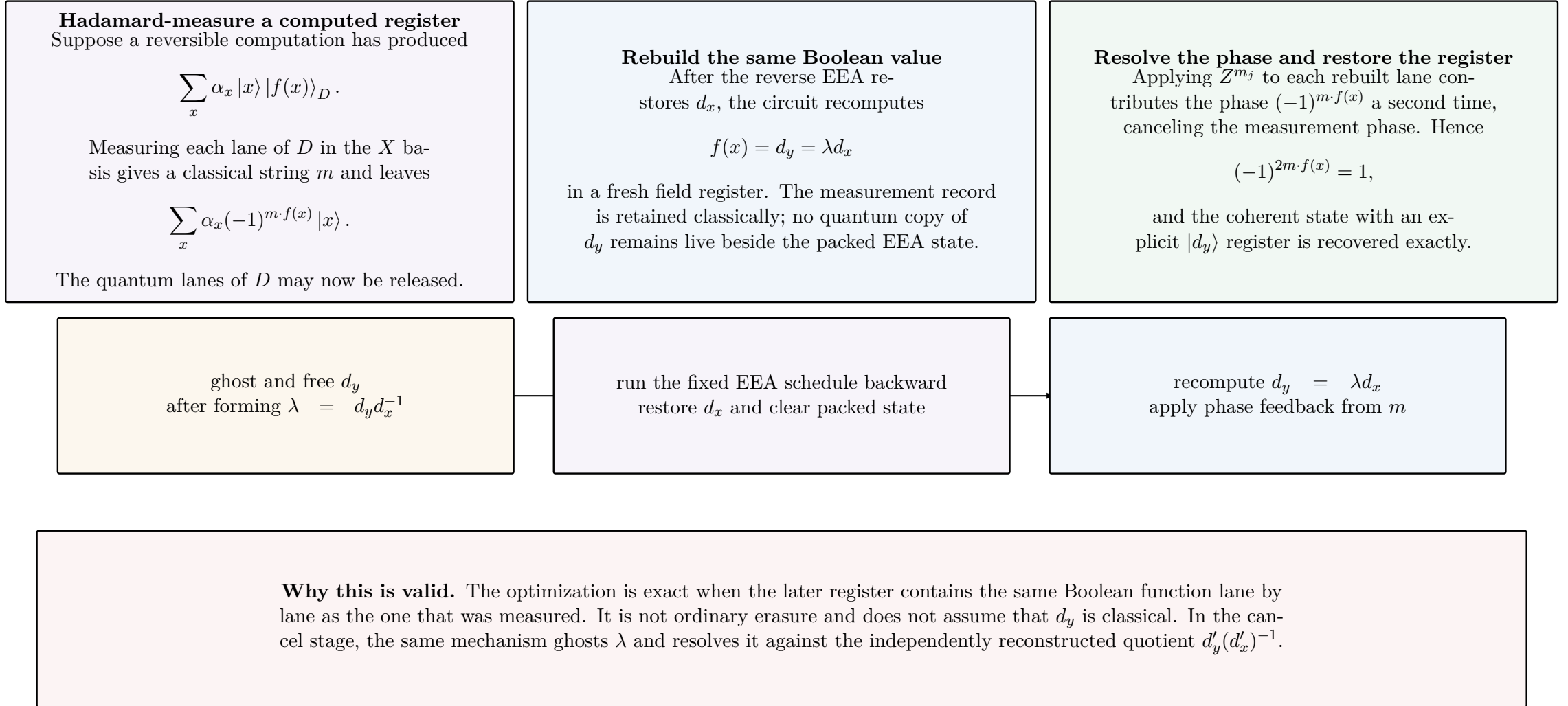
and resolves the stored phase corrections against the reconstructed register.

Top-lane hosting

Canonical field values carry a clean 257th lane. During selected reverse-schedule substages, the clean λ or d_y top lane hosts omitted metadata or predicate scratch, then returns to zero before the modular multiply or API boundary.

Stage 2: measurement-based release and later recomputation

The route frees a full field register during the reverse EEA without discarding its coherent information.



Stage 2: inherited low- Q optimization stack

Q825 composes many earlier width reductions; the omitted ℓ_q lane is only the final step.

Three Q826 host windows

Separate substages lend inactive t' or ℓ_s lanes to remainder, coefficient, and rotated-swap microkernels. Their lifetimes are disjoint, so all three one-qubit savings can coexist.

Metadata relocation

Direct swap metadata, a relocated coefficient counter, a fixed parity bit for ℓ_s , and reuse of ℓ_q as old-remainder metadata reduce the persistent control-register footprint.

Predicate and comparison narrowing

Prefix bit-length circuits operate only on active ranges. Truncated guards, direct selectors, and borrowed underflow lanes avoid full-width comparators when the route supplies a tighter bound.

Paired source complements

Adjacent bit-length kernels retain a complemented source across a source-disjoint use block, allowing the middle pair of X layers to commute and cancel instead of being emitted twice.

Carry and scratch reuse

Zero carries, comparator chains, raw-prefix lanes, the preserved passenger top bit, and clean coefficient scratch are reused across nonoverlapping local kernels and restored at each ownership boundary.

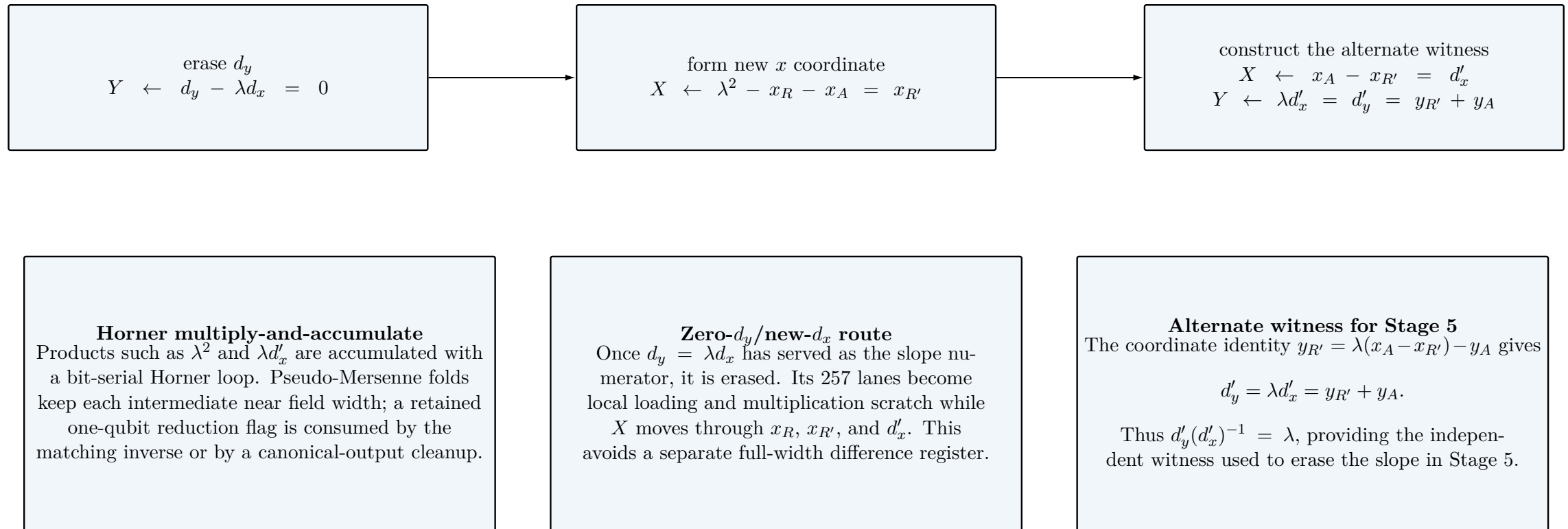
Measurement-assisted cleanup

Temporary logical-AND and carry ancillas are measured in the X basis and corrected by classical phase feedback, avoiding selected reverse-Toffoli cleanup while releasing qubits promptly.

Design principle: the route minimizes the union of simultaneously live information, not the sum of nominal register widths. Most individual changes matter only when they relieve a co-binding peak.

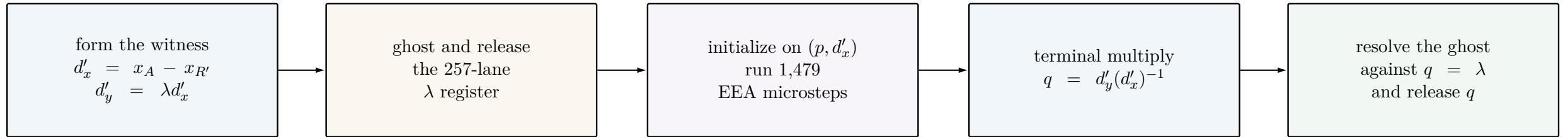
Stages 3–4: coordinate update and alternate-witness construction

The same X and Y registers are recycled from input coordinates to output coordinates.



Stage 5: cancel λ with a fresh alternate witness

The second 825-qubit peak proves and exploits $d'_y(d'_x)^{-1} = \lambda$ for the generic addition case.



Elliptic-curve identity

For $R' = R + A$ in the generic affine case,

$$d'_x = x_A - x_{R'}, \quad d'_y = y_{R'} + y_A = \lambda d'_x.$$

Therefore $d'_y(d'_x)^{-1} = \lambda$ whenever $d'_x \neq 0$.

No stored path from Stage 2

The implementation does not retain or replay the first EEA trajectory. It constructs a new packed EEA state from d'_x , follows the data-dependent schedule for that witness, and produces a fresh quotient whose lanes match the ghosted slope.

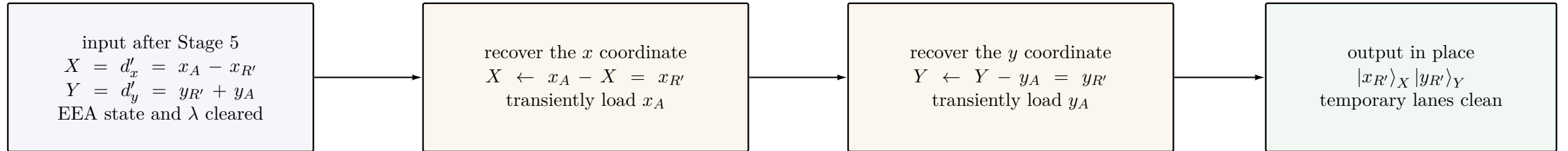
Reversible teardown

After resolving the λ ghost, the terminal multiplication is undone, terminal constants are rebuilt, and the fixed EEA schedule is reversed. This restores d'_x , clears both packed 259-lane banks and all meta-data, and leaves only the output-coordinate route.

The cancellation identity is algebraically exact for the generic addition branch. The submitted circuit's finite-width and truncated arithmetic choices retain the benchmark-support evidential scope stated elsewhere in this deck.

Stage 6: recover the output and clear remaining workspace

The final off-peak stage converts the alternate witness into the affine coordinates of $R' = R + A$.



Algebraic recovery The alternate witness satisfies

$$d'_x = x_A - x_{R'}, \quad d'_y = \lambda d'_x = y_{R'} + y_A.$$

The two subtractions therefore recover $x_{R'}$ and $y_{R'}$ without allocating separate output registers.

Transient classical-coordinate access

The classically known coordinates of A are loaded only for their corresponding subtraction and are removed immediately afterward. The same X and Y lanes that held the Euclidean witnesses become the final output registers.

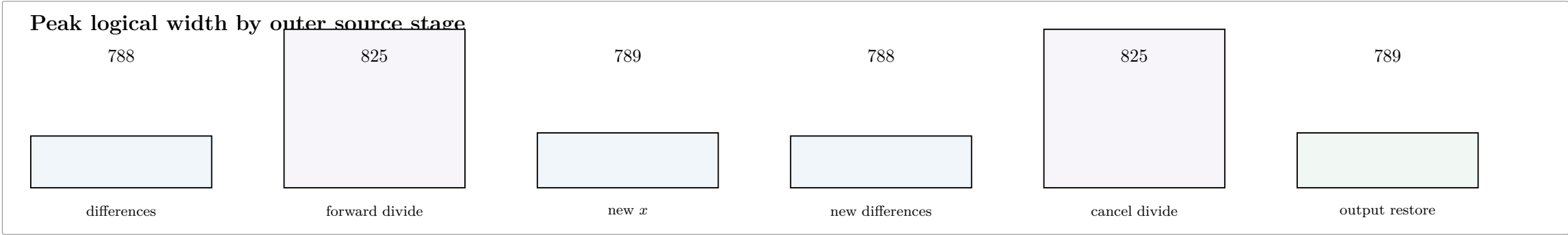
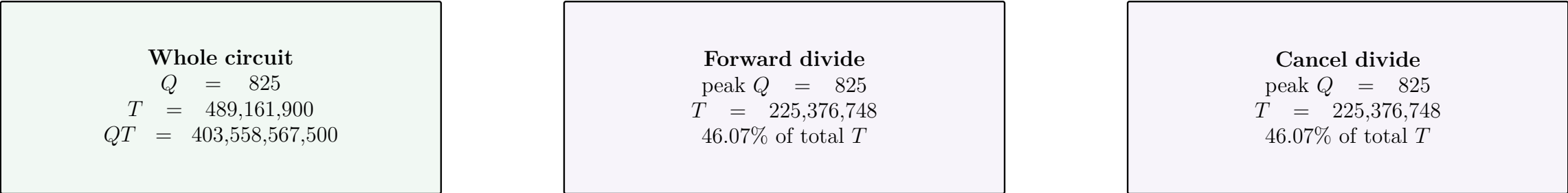
Off-peak reversible teardown

Stage 6 reaches $Q = 789$, below the two $Q = 825$ EEA peaks. The transient load, carry, and padding lanes are returned clean, leaving only the API-width affine coordinates of R' .

Source correspondence: `ec3.alt.new_x_restore` and `ec3.alt.new_y_restore`.

Measured resource profile of the restored source

Count-only instrumentation reproduces the submission and explains the low- Q /high- T tradeoff.



Inversion-dominated cost. The two EEA stages contribute 450,753,496 Toffolis, or 92.15% of the circuit total. The route is therefore a hard-width witness rather than a product-score-efficient operating point.

Marginal Q825 trade. Relative to the cited Q826 parent point (826, 479,337,428), Q825 saves one qubit while adding 9,824,472 Toffolis (2.05%), increasing $Q \times T$ by approximately 1.93%.

Validation record and evidential boundaries

The source contains substantial component and composition checks, but the strongest claims remain finite-support.

Reported evaluator replay

9,024 Fiat–Shamir test cases
zero classical-output failures
zero phase failures
zero ancilla-garbage failures
maximum observed $\ell_q = 22$

Q825 component checks

Ten-control one-short MCX exhaustively checked over 262,144 basis states. The bounded split-five component check covers source lengths $0 \dots 259$, representative source classes, boundary forms, control values, phase, inverse, and scratch restoration.

Composition evidence

The submission note reports Q826/Q825 comparison at 425,952 internal boundaries, 425,984 semantic snapshots, and 64 bit-sliced measurement branches per factor, together with forward/inverse cleanup checks.

What is established

A concrete gate stream with the reported Q and structural T ; clean behavior on the pinned validation corpus; source-level mechanisms for the omitted lane; and bounded verification of the relevant microkernels and selected compositions.

What remains separate

Universal correctness of every support-tuned width promise; completeness of the public archive; coherent QROM compatibility; full windowed-Shor cost; physical error correction; and a proof that the 825-qubit point is globally optimal.

The public status “rejected” should not be read as a correctness diagnosis. In the challenge workflow, a clean but non-promoted hard- Q point may remain rejected because it does not improve the primary leaderboard score.

Source map and reproduction notes

The analysis is tied to submission b6f2b0a and source commit e6d9fa9.

Point-addition driver

src/point_add/trailmix_port/ec/point_add.rs
ec_add_inplace_shrunken_pz

Builds d_x, d_y , invokes the register-shared divider, recycles X, Y through the coordinate formulas, invokes alternate-witness cancellation, and restores the 256-bit interface.

Register-shared EEA

src/point_add/trailmix_port/inversion/
register_shared_eea.rs
register_shared_eea_reference.rs

Defines the packed state, scheduled forward/reverse microsteps, host windows, Q825 implicit-bit kernel, and component/composition checks.

Route configuration and instrumentation

src/point_add/trailmix_port/mod.rs

Source-bakes the low- Q flags and tail nonce, allocates the benchmark registers, reports support statistics, and provides count-only stage/peak instrumentation.

Arithmetic substrate

trailmix_port/rfold_mbu.rs and trailmix_port/arith/

Implements canonical and rfold modular multiplication, pseudo-Mersenne correction, comparison, prefix, multi-control, carry, and measurement-assisted cleanup primitives.

Restore with ecdsafail reset b6f2b0a. For a non-emitting structural recount, run the release builder with POINT_ADD_COUNT_ONLY=1, TRACE_PHASE_OPS=1, TRACE_PHASE_ACTIVE=1, and TRACE_STRUCTURAL_COUNT=1.

Primary references for the architecture: Proos–Zalka reversible ECDLP arithmetic and the later register-sharing formulation cited by the source; the production implementation is the audited challenge artifact above.